NAME

DVIasm – a TeX utility program for editing DVI files directly

SYNOPSIS

dviasm [<*options*>] <*dvi_file*>|<*dvi_dump_file*>

DESCRIPTION

DVIasm is a TeX utility program which is designed for editing DeVice-Independent (DVI) files directly. It consists of a single Python script, dviasm.py, in a human readable text format. It runs on any platform in which Python 3 is installed.

Features of **DVIasm** includes:

1) Disassemble a DVI file (or XeTeX XDV file) into a human-readable text file, the contents of which are easy to modify.

2) Assembles the output text file back to the binary format.

There are several alternatives for dumping and editing DVI files:

1) **DVItype** developed by Donald E. Knuth supports one-way conversion from DVI to a text format. There are also some derivatives: **pDVItype** for pTeX, **upDVItype** for upTeX, **ODVItype** for Omega.

2) The programs **dv2dt** and **dt2dv** can be used in pairs to allow two-way conversion between DVI and the DVI Text Language (DTL).

3) The program **dvispc** (part of DVIOUT previewer) also has an ability to convert between DVI and text.

Among those, **DVIasm** is designed to allow users to edit DVI files easily, for example by unifying ‘‘right1’’–‘‘right4’’ to a single command ‘‘right’’ which can be used regardless of the amount of move.

OPTIONS

--version Show program’s version number and exit.

-h, **--help** Show this help message and exit.

-u <*STR*>, **--unit=**<*STR*>
Set unit [default: ‘pt’]. Allowed values are: ‘sp’, ‘pt’, ‘bp’, ‘mm’, ‘cm’ and ‘in’.

-o <*FILE*>, **--output=**<*FILE*>
Set filename for output instead of stdout.

-e <*STR*>, **--encoding=**<*STR*>
Set encoding for input/output of dumped text [default: ‘utf8’]. Allowed values are: ‘ascii’, ‘latin1’, ‘utf8’, ‘sjis’ and ‘eucjp’. When used with the option **-p** (or **--ptex**), allowed values are only ‘utf8’, ‘sjis’ and ‘eucjp’.
NOTE: This feature does not support dumping through stdout.

-x <*STR*>, **--xxx-encoding=**<*STR*>
Set encoding for interpreting ‘xxx:’ strings [default: ‘none’]. Allowed values are: ‘none’, ‘utf8’, ‘sjis’ and ‘eucjp’.
This option allows users to specify the correct encoding for dumping/compiling the contents of \special.

Typical usages are as follows:

1) By default, the contents of \special are dumped/compiled as byte-to-byte escape sequences in '\x..' format. This would suffice when you don't need to read/edit the contents of \special.

3) For XeTeX and upTeX users, '**-x utf8**' would be useful since these engines use UTF-8 to encode strings inside \special commands.

2) For pTeX users with ISO-2022-JP-encoded DVI, it would be helpful to select the appropriate option '**-x eucjp**' (for Unix) or '**-x sjis**' (for Windows) to be consistent with the internal Kanji encoding used by pTeX engine.

-t <*INT*>, **--tabsize=**<*INT*>
Set tab size for push/pop [default: 2].

-p, **--ptex** Dump and compile ISO-2022-JP-encoded DVI for Japanese pTeX. This option is required to decode or encode Japanese characters being typeset (using 'set2' or 'put2') properly.

-s <*STR*>, **--subfont=**<*STR*>
Specify the list of fonts with UCS2 subfont scheme in the comma-separated format.
If <*STR*> is empty, the pre-defined internal subfont list is disabled.
This option would be useful for example when using CJK package.

EXAMPLE

Documentation of **DVIasm** by the original author Jin-Hwan Cho is published as:
http://tug.org/TUGboat/Articles/tb28-2/tb89cho.pdf (TUGboat)
http://ajt.ktug.kr/assets/2008/5/1/0201cho.pdf (Asian Journal of TeX)

AUTHORS

The first version was written in Python 2 by Jin-Hwan Cho in 2007. Khaled Hosny started to support dumping/compiling XeTeX XDV format in 2014. Arthur Reutenauer ported the script to Python 3 in 2019. Current maintainer of **DVIasm** is Hironobu Yamashita, since 2020.

BUG REPORT

Please visit the GitHub repository <https://github.com/aminophen/dviasm>.

SEE ALSO

dvitype(1), **dv2dt**(1), **dt2dv**(1), **dvispc**(1).